Multiplication in $\mathbb{Q}$

1. $\frac{3}{4} \times \frac{5}{7}$
2. $\frac{4}{9} \times \frac{3}{8}$
3. $-\frac{5}{12} \times \frac{4}{15}$
4. $\left(-\frac{7}{10}\right) \times \left(-\frac{5}{14}\right)$
5. $1\frac{3}{5} \times 2\frac{1}{2}$
6. $-3\frac{1}{3} \times 1\frac{1}{5}$
7. $\frac{2}{3} \times \left(-\frac{9}{10}\right) \times \frac{5}{8}$
8. $\left(-2\frac{1}{4}\right) \times \left(-1\frac{7}{9}\right)$
9. A water tank holds 45 liters. If the tank is currently $\frac{3}{5}$ full, how many liters of water are in the tank?
10. A rectangular plot of land measures $\frac{4}{5}$ kilometers long and $\frac{3}{8}$ kilometers wide. What is the area of the plot of land in square kilometers?

Solutions

1. $\dfrac{15}{28}$

   Explanation: Multiply numerators together and denominators together:

   $$\frac{3 \times 5}{4 \times 7} = \frac{15}{28}.$$

2. $\dfrac{1}{6}$

   Explanation: Multiply and simplify by canceling common factors:

   $$\frac{4}{9} \times \frac{3}{8} = \frac{4 \times 3}{9 \times 8} = \frac{12}{72}.$$

   Divide numerator and denominator by 12: $\dfrac{12 \div 12}{72 \div 12} = \dfrac{1}{6}$.

3. $-\dfrac{1}{9}$

   Explanation: Product of negative and positive is negative. Simplify common factors (4 with 12, 5 with 15):

   $$-\left(\frac{5}{12} \times \frac{4}{15}\right) = -\left(\frac{1}{3} \times \frac{1}{3}\right) = -\frac{1}{9}.$$

4. $\dfrac{1}{4}$

   Explanation: Product of two negative numbers is positive. Simplify common factors (7 with 14, 5 with 10):

   $$\left(-\frac{7}{10}\right) \times \left(-\frac{5}{14}\right) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

5. 4

   Explanation: Convert mixed numbers to improper fractions:

   $$1\frac{3}{5} = \frac{8}{5}, \quad 2\frac{1}{2} = \frac{5}{2}.$$

   Multiply and simplify:

   $$\frac{8}{5} \times \frac{5}{2} = \frac{8 \times 5}{5 \times 2} = \frac{40}{10} = 4.$$

6. -4

   Explanation: Convert to improper fractions:

   $$-3\frac{1}{3} = -\frac{10}{3}, \quad 1\frac{1}{5} = \frac{6}{5}.$$

   Multiply:

   $$-\left(\frac{10}{3} \times \frac{6}{5}\right) = -\left(\frac{60}{15}\right) = -4.$$

7. $-\dfrac{3}{8}$

   Explanation: Combine terms and simplify:

$$\frac{2}{3} \times \left(-\frac{9}{10}\right) \times \frac{5}{8} = -\left(\frac{2 \times 9 \times 5}{3 \times 10 \times 8}\right) = -\left(\frac{90}{240}\right) = -\frac{3}{8}.$$

8. 4

   Explanation: Negative times negative is positive. Convert to improper fractions:

$$-2\frac{1}{4} = -\frac{9}{4}, \quad -1\frac{7}{9} = -\frac{16}{9}.$$

   Multiply and cancel common factors:

$$\frac{9}{4} \times \frac{16}{9} = \frac{1}{1} \times \frac{4}{1} = 4.$$

9. 27 **liters**

   Explanation: Calculate fraction of whole quantity:

$$45 \times \frac{3}{5} = \frac{45 \times 3}{5} = \frac{135}{5} = 27 \text{ liters}.$$

10. $\dfrac{3}{10}$ $\mathbf{km}^2$

    Explanation: Area = length × width:

$$\text{Area} = \frac{4}{5} \times \frac{3}{8} = \frac{4 \times 3}{5 \times 8} = \frac{12}{40} = \frac{3}{10} \text{ km}^2.$$